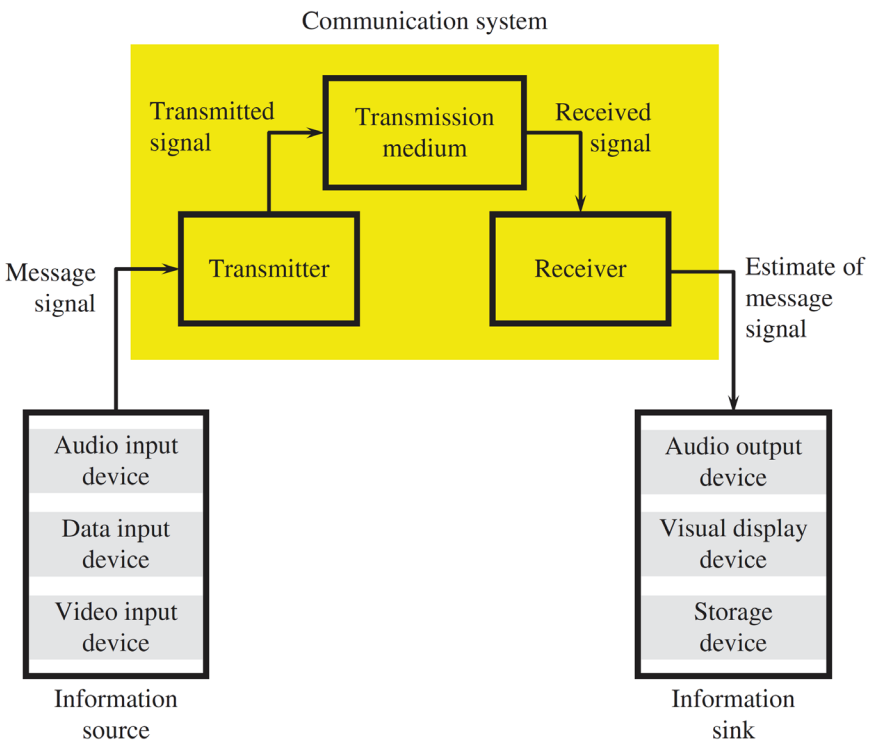
Question 1

- What are the various transducers used in communication systems?
- A transducer is a device that converts variations in a physical quantity, into an electrical signal.
- Microphones, loudspeakers, TV camera (to convert light into electrical signal) etc. are some of the transducers used in communication systems.

传感器：将信息信号转换成电信号



Question 2

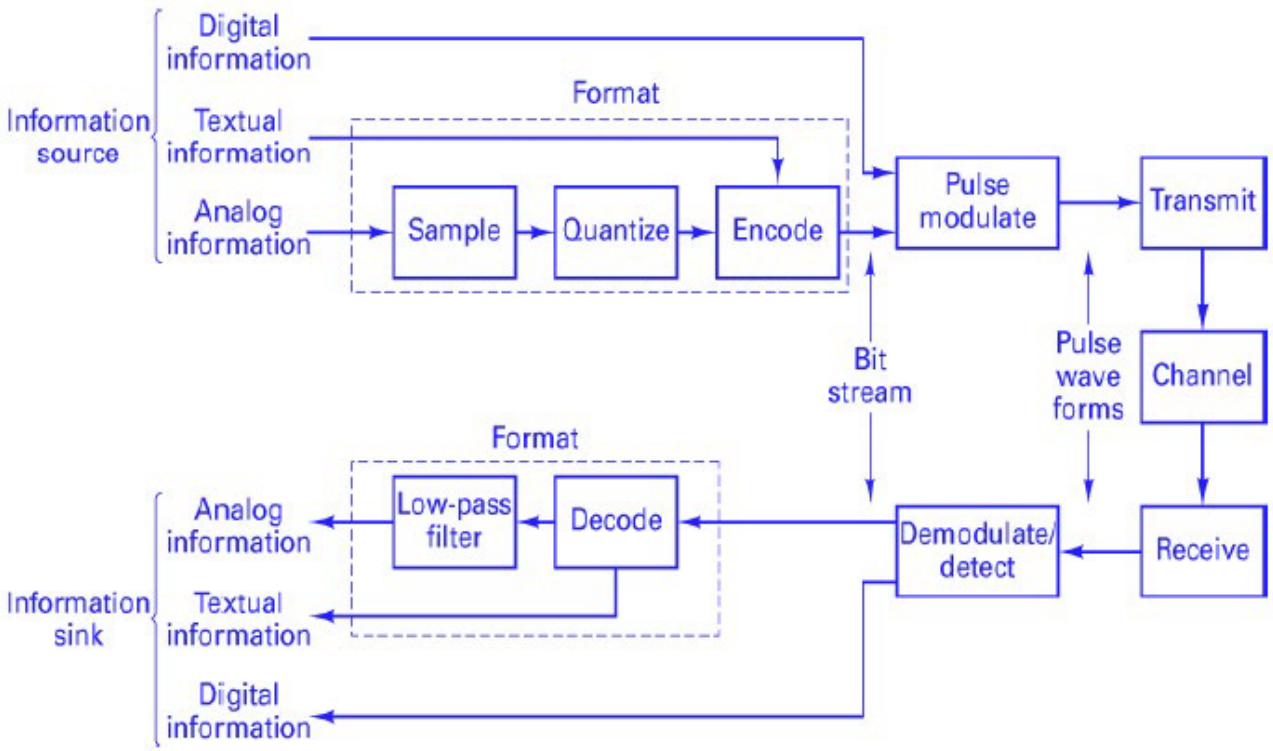
- What do you understand by the term baseband signal?

- The electrical equivalent of the original message signal (i.e. sound, picture, computer data) is called as baseband signal.

原始信息信号的电子等效

Question 3

- Source Coding
- The job of source coding is to remove as much redundancy from the message as possible to shorten the digital message sequence that requires transmission. Source coding aims to use as little bandwidth as possible without considering channel noise and interference.



Question 4

Characterise the main areas of image processing: image compression, image enhancement and image understanding. State what their goals are and give examples.

- **Image processing** is a subclass of signal processing specifically concerned with pictures. Image processing has three main areas:
  - Image compression
  - Image enhancement and
  - Image understanding
- **Image compression** is concerned with how we can reduce the redundancy of the image data in order to be able to store or transmit data in an efficient form (lossy or lossless).

Cont'd

- **Image enhancement:** aims to improve perception of information in images for human viewers, or to provide `better' input for other automated image processing techniques. An example is the reduction of noise in an image and revealing of particular features in the image. Enhancement alters an image to makes its meaning clearer to human observers.
- **Image understanding:** the process of interpreting the regions, objects in the image, to figure out what actually happens in the image. This may be figuring out what the objects are, how many objects there are, their spatial relationships to each other, etc. It may also include ultimately making decisions for further actions. The goal is to understand the 'content' of an image or sequence of images (video) and to extract meaning of the image (content based retrieval). The process can be performed bottom-up or top-bottom.

Image Processing Example: Filtering

- Filter an image by replacing each pixel in the source with a weighted sum of its neighbors
- Define the filter using a *convolution mask*, also referred to as a *kernel*
  - non-zero values in small neighborhood, typically centered around a central pixel
  - generally have odd number of rows/columns

Convolution Filter

100	100	100	100	100
100	100	50	50	100
100	100	100	100	100
100	100	100	100	100
100	100	100	100	100

X

	0	1	0	
	0	0	0	
	0	0	0	

=

100	100	100	100	100
100	100	50	50	100
100	100	50	100	100
100	100	100	100	100
100	100	100	100	100

Mean Filter

20	12	14	23
45	15	19	33
55	34	81	22
8	64	49	95

Subset of image

$$\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Convolution filter

output

$(20+12+14+45+15+19+55+34+81)/9$

Common 3x3 Filters

- Low/High pass filter

$$\frac{1}{9}\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix} \qquad \begin{bmatrix}-1 & -1 & -1\\ -1 & 9 & -1\\ -1 & -1 & -1\end{bmatrix}$$

- Blur operator

$$\frac{1}{13}\begin{bmatrix}1 & 2 & 1\\ 2 & 1 & 2\\ 1 & 2 & 1\end{bmatrix}$$

- H/V Edge detector

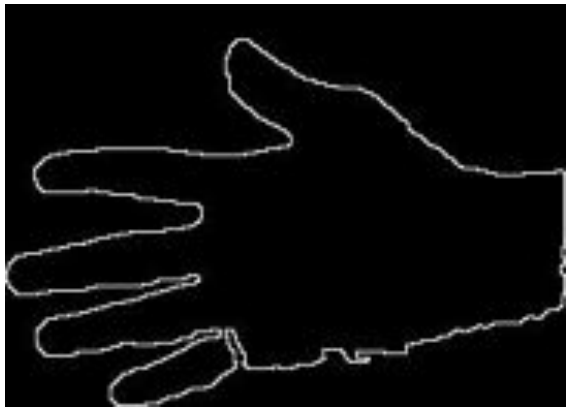
$$\begin{bmatrix}1 & 2 & 1\\ 0 & 0 & 0\\ -1 & -2 & -1\end{bmatrix} \qquad \begin{bmatrix}-1 & 0 & 1\\ -2 & 0 & 2\\ -1 & 0 & 1\end{bmatrix}$$

Low and High Pass filter example

Image Processing example: Edge Detection

- Identify areas of strong intensity contrast
  - filter useless data; preserve important properties

- Other technique
  - identify shapes, match to templates, invoke commands



Simple Edge Detection

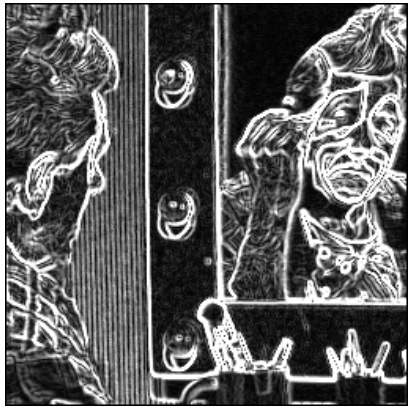
- Example: Let assume single line of pixels

5	7	6	4	152	148	149

- Calculate 1st derivative (gradient) of the intensity of the original data
  - **Using gradient, we can find peak pixels in image**
  - *I(x)* represents intensity of pixel *x* and
  - *I'(x)* represents gradient (in 1D),
  - Then the gradient can be calculated by convolving the original data with a mask (-1/2 0 +1/2)
  - I'(x) = -1/2 *I(x-1) + 0*I(x) + ½*I(x+1)

Basic Method of Edge Detection

- Step 1: filter noise using mean filter
- Step 2: compute spatial gradient
- Step 3: mark points > *threshold* as edges



Question 5

- Explain the difference between the high- and low-level image processing tasks.
- **Low level**: input, output are images

  This level involves primitive operations such as image pre-processing to reduce noise, image enhancement, and image sharpening. There is **no interpretation** involved at the level of pure image processing operations, such as brightness, and contrast enhancement.

- **Mid level**: input may be images, outputs are attributes extracted by the images

  Image segmentation (partitioning an image into regions)

  Description of objects

  Classification of objects

Cont'd

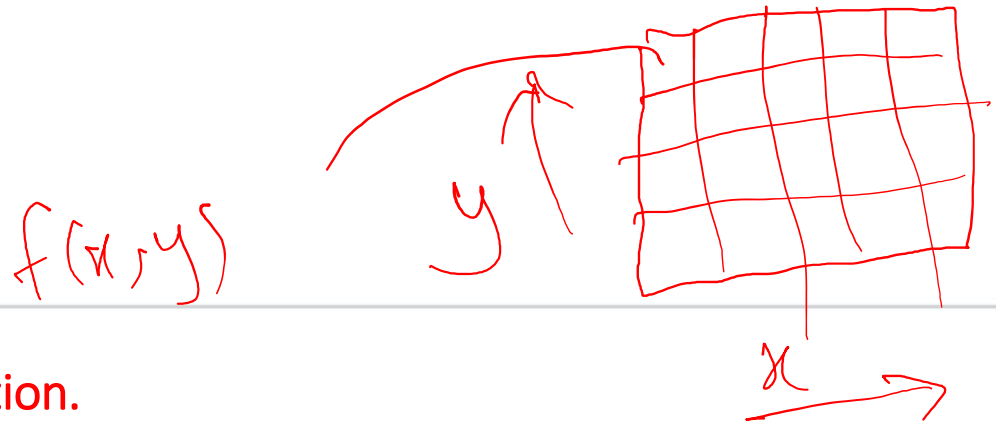
- **High level**: image analysis, cognitive functions related with the human vision

  Interpreted to an application specific domain.

  Artificial intelligence techniques

  Often need to be simplified.

Question 6



Explain the digital image representation.

- A digital image is a 2-D array with numbers, roughly speaking. For binary and grey scale images we have one array only. For colour images, such as RGB and HSV, we have three arrays (matrices) for each colour.
- **An image:** can be defined as a 2-D light intensity function f(x,y) , where x and y denotes spatial co-ordinates and the value of *'f'* at any point is directly proportional to the intensity (or) grey level of the image at that point.

- Categories of images by sources: optical, infrared (IR), satellite, ultrasound, etc.

Question 7
The Human Eye vs the Camera

Camera

- Imaging surface: rectangle except panoramic cameras
- Lens, Sensor: CCD e.g.
- Percise, but with no interpretation
- **Light** is transformed into **electrical signals** and converted into **images/ video**.

Human Vision/Eye

- Imaging surface: spherical
- Lens
- Retina: with photosensitive elements, rods and cones
- **Uses former experience in the process of interpretation**
- **Light** penetrating the eye is transformed into **neural impulses** via the optical nerve to the brain where the **image** is formed.

Eye vs. Camera

Camera components	Eye components
Lens	Lens, cornea
Shutter	Iris, pupil
Film	Retina
Cable to transfer images	Optic nerve send the info to the brain

Question 8

- Optical camera vs infrared (thermal) camera.
- The optical camera converts light into electrical impulses and then into digital form (numbers). The thermal (infrared camera) converts the emitted infrared radiation from objects into temperature maps.

- The IR camera detects and measures IR energy emitted from objects, convert the data to corresponding maps of temperatures, and displays an image of temperature distribution. The temperature can be measured from a distance without contacting an object.

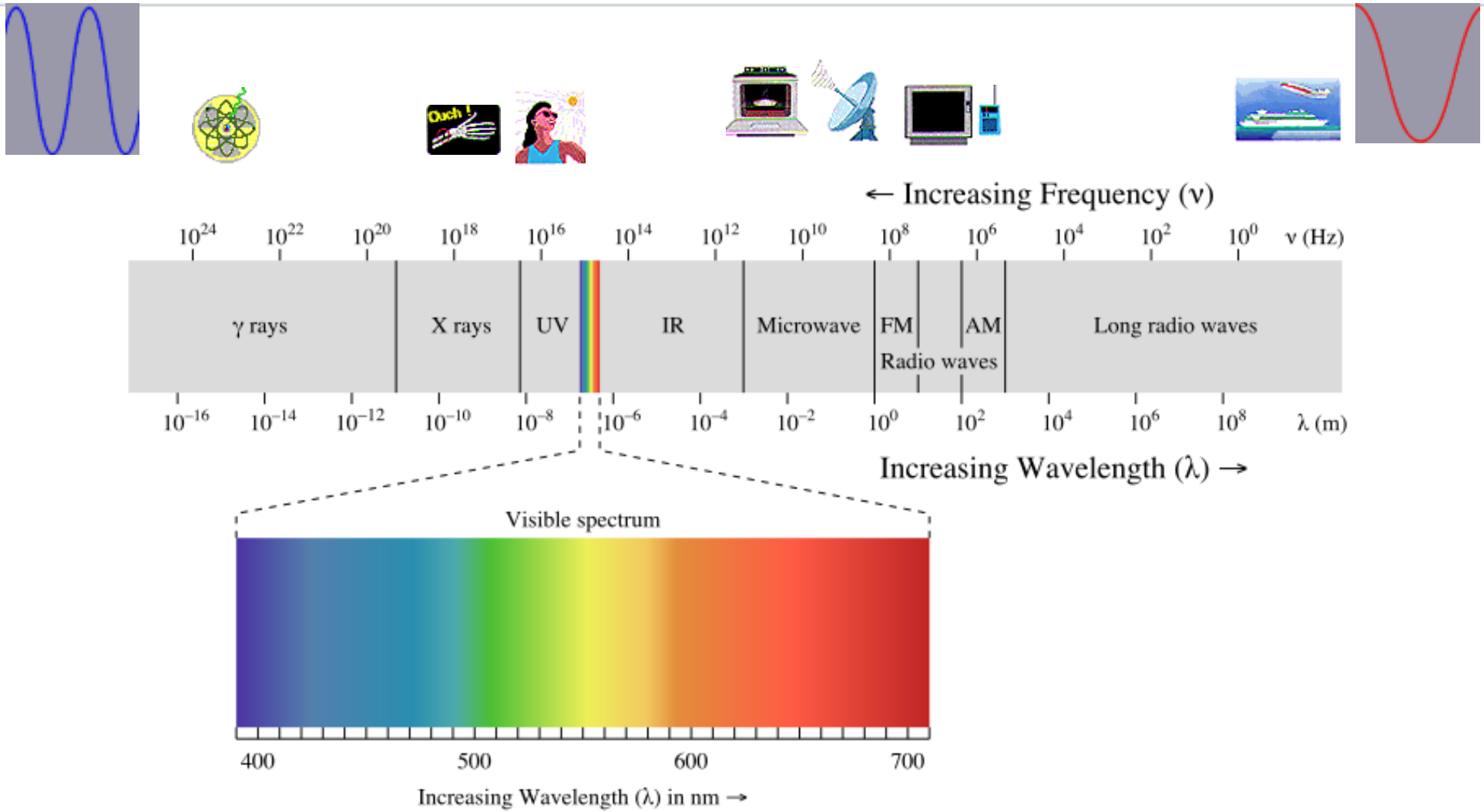
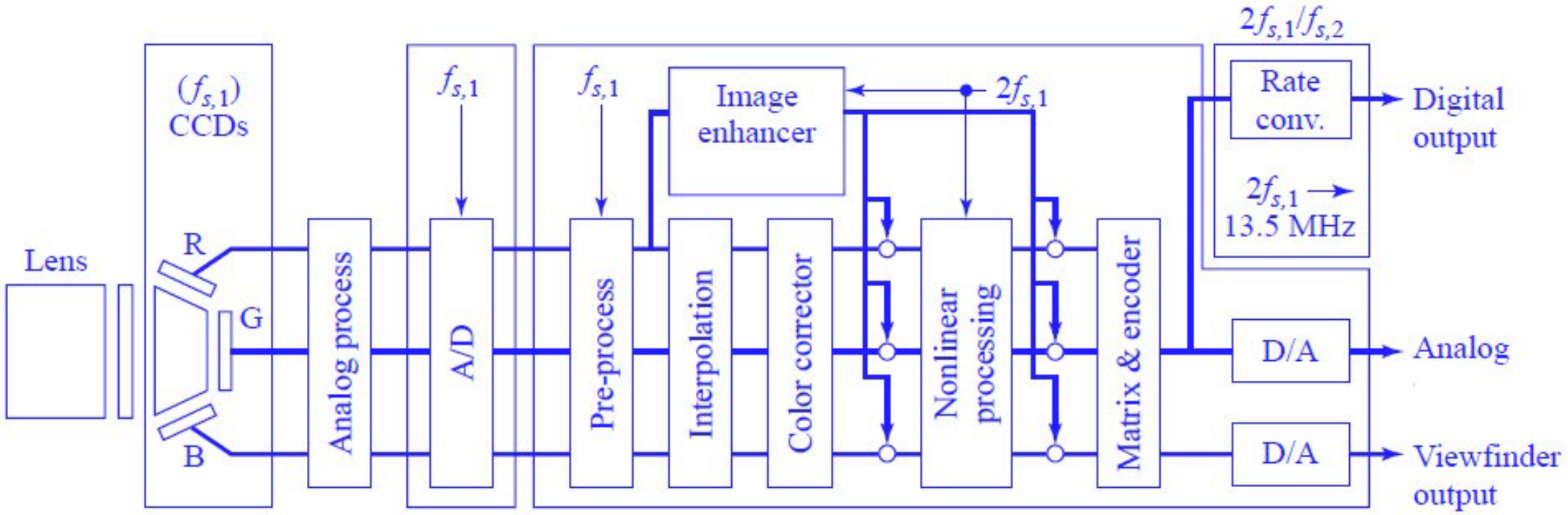
Question 9

Why are the infrared (IR) rays able to penetrate given medium rather easily?

- Unlike the visible rays, infrared rays tend to penetrate any medium rather easily because of their large wavelengths.

- Information from IR and visual images can be combined: allows seeing the invisible.

EM Spectrum


← Increasing Frequency (ν)
10^{24} 10^{22} 10^{20} 10^{18} 10^{16} 10^{14} 10^{12} 10^{10} 10^{8} 10^{6} 10^{4} 10^{2} 10^{0} ν (Hz)
γ rays
X rays
UV
IR
Microwave
FM
AM
Radio waves
Long radio waves
10^{-16} 10^{-14} 10^{-12} 10^{-10} 10^{-8} 10^{-6} 10^{-4} 10^{-2} 10^{0} 10^{2} 10^{4} 10^{6} 10^{8} λ (m)
Increasing Wavelength (λ) →
Visible spectrum
400
500
600
700
Increasing Wavelength (λ) in nm →

Professional Colour Video Camera